

Chapter 160 — Open Radical Prostatectomy — Numbers, Gold Standards and Traps

IF YOU READ NOTHING ELSE

- **Radical prostatectomy (RP)**: only treatment with randomised controlled trial (RCT) proof ↓ metastases and death (SPCG-4)
- **Goals in order**: cancer → complications → continence → potency
- **Defer surgery**: 6 wk after biopsy, 12 wk after transurethral resection of the prostate (TURP)
- **Commonest positive margin site**: apex → posterior → posterolateral
- **Interfascial dissection** = correct plane; **intrafascial** never
- **Neurovascular bundle (NVB)**: clips, never thermal energy
- **Induration or fixed bundle** → excise NVB
- **Hypotension needing transfusion** → explore and evacuate haematoma
- **Deep vein thrombosis (DVT) or pulmonary embolism (PE)** = main cause of death; peak 14–28 days
- **Rectal injury** → 2-layer repair + omental interposition; colostomy if irradiated

OPENING

- **Robotic RP** dominant; **no proven** oncological or functional superiority over open
- **Main robotic advantage**: less blood loss only
- **Man 65–70 y**: 50% chance of living 15 more years

SURGICAL ANATOMY

- **Deep dorsal vein** → 3 branches: superficial + right and left lateral plexuses
- **Superficial branch**: outside anterior prostatic fascia, between puboprostatic ligaments
- **Dorsal vein transected at common trunk**, directly over urethra
- **Prostatic artery**: inferior vesical → **urethral** (periurethral) + **capsular** (outer gland) groups
- **Capsular vessels** = macroscopic landmark for cavernous nerves (NVB of Walsh)
- **Cavernous artery source**: internal pudendal; **accessory pudendals** from obturator or vesical arteries
- **Pelvic plexus**: parasympathetic S2–S4 + sympathetic via hypogastric nerve
- **Plexus position**: 5–11 cm from anal verge; midpoint at **seminal vesicle tip**
- **Mid-pedicle ligation** cuts vessels and nerves
- **NVB forms 20–30 mm** distal to prostatovesical junction
- **NVB plane**: between **levator fascia and prostatic fascia**
- **Nerves on the capsule** enter the prostate, not the corpora
- **Striated sphincter** = **vertical tube** (Oelrich), not horizontal "sandwich"
- **Passive continence** = striated sphincter (slow-twitch) **vs active** = levator ani; both **pudendal nerve**
- **Denonvilliers**: layers inseparable → **excise completely**
- **Perineal approach** reflects fascia → less bleeding **vs retropubic**: DVC must be ligated

SURGICAL TECHNIQUE

Preoperative:

- **Coexisting inguinal hernia** ~15%; **after open RP up to 20%** → examine standing with Valsalva; repair preperitoneally at RP
- **No autologous donation**; erythropoietic agents ↑ haematocrit
- **Prophylaxis**: 1st- or 2nd-generation cephalosporin, **or** aminoglycoside + metronidazole or clindamycin
- **Coronary stent**: low-dose aspirin usually continued; plan with cardiologist
- **Magnetic resonance imaging (MRI)** guides NVB and membranous urethra preservation

Gold standard / essentials:

- **Fibreoptic headlight**: essential; **Balfour retractor** required

- **General endotracheal anaesthesia**; systolic ≤100 mm Hg; crystalloid ≤1500 mL until prostate out
- **16-Fr silicone Foley** (eases mucosal suturing); right-hander stands on left

Lymphadenectomy:

- **Pelvic lymph node dissection (PLND)** for intermediate or high risk; **omit for Grade Group 1–2** low-volume favourable
- **Spare tissue over external iliac artery** → avoids leg oedema and lymphocele
- **Node of Cloquet**: removal unnecessary
- **Extended PLND** → better staging ± therapeutic benefit

Apex and dorsal vein complex (DVC):

- **Endopelvic fascia**: incise at sidewall reflection; too medial → Santorini bleeding
- **Pudendal perforators**: clip, not coagulate
- **Spare pubourethral ligament component** → anterior sphincter fixation
- **Accessory pudendal**: cadaver ~70% vs angiography ~7%; large ~4%
- **Unilateral accessory artery** → **do contralateral side first**
- **Stapler on DVC**: fire only if **Foley moves freely**; ~200–400 mL less blood loss
- **Suture ligation**: 3–0 through DVC + pubic perichondrium (figure-of-eight)
- **Divide sphincter edges obliquely midway apex** ↔ pelvic floor (NVB tethered at apex)
- **Circumflex veins at 5 and 7 o'clock** → clips

Urethra:

- **Standard release**: urethra divided **first**; **high release**: urethra divided **after**
- **6 × 3-0 Monocryl**: **mucosa + submucosa only**; smooth muscle bites delay continence
- **Posterior sphincter**: clamp midway, **left then right then centre**; never all from one side

NVB:

- **Most men**: both NVBs preserved; **bilateral excision almost never**
- **Extracapsular extension** rarely >1–2 mm
- **Margin rate same** nerve-sparing vs non-nerve-sparing; Walsh overall ~5%
- **Final NVB decision**: **intraoperative**, never preoperative
- **Release apex** → **base**, from rectal surface; roll prostate, no upward traction
- **Posterior branch over seminal vesicle** → ligate → NVB falls away
- **Tsuzuki**: ≥2 of prostate-specific antigen (PSA) ≥2 of PSA >10gt; 10, Gleason >6, core >20%, cores >33%, perineural invasion, abnormal digital rectal examination → >10% extension risk

- **High anterior release:** after DVC ligation, **before urethra**; unilateral = bilateral → do one side
- **Wide excision:** free contralateral NVB first; clamp **medial** → **lateral** on rectum

Pedicles, bladder neck, anastomosis:

- **Lateral pedicle:** 3 layers (superficial, middle, deep)
- **Sponge stick on catheter**, never on prostate
- **Pedicle branches to bladder neck at 5 and 7 o'clock**
- **Pelvic plexus lateral to seminal vesicle** → hug seminal vesicle
- **Complete seminal vesicle excision** essential for cancer control
- **Doubtful posterolateral margin on specimen** → **excise that NVB**
- **Tennis-racquet closure** 2-0; mucosal eversion 4-0 → mucosa-to-mucosa
- **Intussusception:** 2-0 Maxon ~2 cm from neck; "turtle sign"
- **Babcock on bladder** while tying (no catheter traction); **anterior suture tied first**
- **Drain between, not through, recti**

Postoperative:

- **Mean stay ~1.7 days**; discharge day 1-2
- **Ketorolac only** if creatinine <1.2 mg/dL, blood loss <1500 mL, no uncontrolled diabetes or haematuria
- **Drain out** <50 mL/day
- **Catheter out at 1 wk:** cystogram, or no imaging if haematocrit fall <6%
- **Delayed anastomotic healing** = perianastomotic haematoma
- **Spasm:** oxybutynin 5 mg or diazepam 5-10 mg

COMPLICATIONS

- **Mortality 0.2%**
- **Commonest intraoperative complication: venous haemorrhage**
- **Troublesome DVC bleed** → **divide completely, oversew**
- **Blood loss 300-1000 mL**; type and screen for all
- **Obturator nerve cut** → **reanastomose**, fine non-absorbable
- **Rectal injury:** apical dissection of **small prostates**
- **Ureteric injury** (trigone dissection) → reimplant
- **Significant bleed** = transfusion needed to support blood pressure
- **DVT risk highest with PLND**; vigilance 4 wk; compression + early ambulation

- **DVT or PE treatment:** twice-daily subcutaneous low-molecular-weight heparin ≥ intravenous unfractionated heparin
- **Bladder neck contracture:** 0.5-10%; presents ~3 months
- **Contracture ladder:** dilatation → cold-knife 3/6/9 o'clock → triamcinolone 200 mg → artificial urinary sphincter
- **Incontinence** = intrinsic sphincter deficiency; main cause **DVC injury**
- **Stop α-blockers** until dry
- **Continence (0-1 pad):** 80 / 90 / 93 / 97% at 3 / 6 / 12 / 24 mo
- **Potency (Walsh):** 38 / 54 / 73 / 86% at 3 / 6 / 12 / 18 mo
- **Potency predictors:** age <65, preoperative potency, no diabetes, bilateral NVB
- **Erectile recovery** continues up to 4 y
- **Padma-Nathan:** nightly sildenafil 50 mg × 9 mo → better function at 1 y
- **REINVENT:** vardenafil on-demand > nightly
- **Other sexual effects:** penile shortening, climacturia
- **Moderate-to-severe International Prostate Symptom Score** → lower urinary tract symptoms improve after RP for up to 15 y

CLASSIC TRAPS

- **Trap:** excise NVB to avoid margins — **Actually:** apex commonest; rates equal
- **Trap:** intrafascial is safer — **Actually:** high margin risk; never used
- **Trap:** nerves on capsule are cavernous — **Actually:** they enter the prostate
- **Trap:** sphincter is a horizontal diaphragm — **Actually:** vertical tube
- **Trap:** catheter out day 4 if no leak — **Actually:** high retention rate
- **Trap:** unstable bleed → observe — **Actually:** explore early
- **Trap:** clamp lateral → medial on rectum — **Actually:** medial → lateral
- **Trap:** nightly phosphodiesterase-5 (PDE5) inhibitor best — **Actually:** on-demand ≥ nightly
- **Trap:** partial DVC division controls bleeding — **Actually:** divide fully
- **Trap:** bilateral high release needed — **Actually:** unilateral equivalent